

ABDULLAH

AI / ML ENGINEER

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PROFESSIONAL SUMMARY

AI/ML Engineer who builds and ships complete, production-style AI systems rather than notebook experiments — from data ingestion and model training to LLM-powered agents deployed behind a live API. Hands-on experience integrating LLM gateways (Mesh API, OpenAI-compatible chat and embedding models) into retrieval-augmented, agentic applications built with LangGraph and vector databases, alongside full MLOps pipelines covering ingestion, transformation, experiment tracking (MLflow/DagsHub), containerization, and cloud deployment (AWS). Comfortable owning a project end to end: designing the architecture, writing the backend (FastAPI), wiring in AI/embedding models, and shipping it as a working web application. Selected among a worldwide pool of participants and ranked **8th place globally** in the SmartReco Hackathon Build Challenge, organized by Krish Naik Academy, for building and deploying a fully functional agentic AI recommendation platform under competition constraints.

TECHNICAL SKILLS

Languages: Python, SQL

AI / ML / LLM: Machine Learning, Deep Learning, NLP, LangGraph (Agentic AI Workflows), Retrieval-Augmented Generation (RAG), Mesh API, OpenAI Chat & Embedding Models, ChromaDB (Vector Database), Scikit-learn, Prompt & Grounding Validation

MLOps & Cloud: MLflow, DagsHub, Docker, GitHub Actions (CI/CD), AWS (EC2, S3, ECR)

Web Development & APIs: FastAPI, REST API Design, SQLAlchemy, Jinja2, MongoDB Atlas, SQLite

Tools & Practices: Git/GitHub, LangSmith (Observability), APScheduler, Agile Problem-Solving

PROJECTS

SmartReco — Behavioral AI Recommendation Agent

★ **8th Place Worldwide** — SmartReco Hackathon (Krish Naik Academy)

Python · FastAPI · LangGraph · ChromaDB · Mesh API (OpenAI-compatible) · SQLAlchemy · APScheduler · LangSmith [View Repository](#)

- **Built an agentic e-learning recommendation platform** from scratch, competing worldwide in a hackathon organized by Krish Naik Academy and finishing 8th globally.
- Designed a real-time behavioral tracking layer that captures views, dwell time, searches, clicks, and scroll depth via a lightweight client-side tracker, batching and flushing events non-blockingly so tracking never slows down the user experience.
- Engineered a multi-factor interest-scoring engine combining recency decay, dwell-time multipliers, frequency boosts, and a category-dominance rule to translate raw behavioral signals into accurate per-category interest scores.
- **Integrated LLM and embedding models through the Mesh API** (OpenAI-compatible gateway) for both narrative generation and semantic embeddings — direct hands-on experience wiring chat and embedding models into a production application.
- Architected a 6-node LangGraph agent (analyze → decide → retrieve → evaluate → refine → generate) that performs category-aware hybrid RAG retrieval against ChromaDB, re-ranks candidates, and generates grounded, persuasive recommendation narratives.
- Built a hallucination-guard layer that validates every generated narrative against real catalog data before it is shown to the user, with automatic fallback if grounding fails twice.
- Implemented a dual-write architecture (SQLite as source of truth + ChromaDB for embeddings) with automatic hourly self-healing reconciliation, so vector-store outages never cause data loss.
- Shipped proactive delivery via APScheduler-driven daily digest emails and Telegram notifications, plus an admin panel to trigger and monitor digests on demand.
- Ran a full security-hardening pass on the platform: closed a privilege-escalation gap on admin routes, fixed a course-ownership spoofing bug, added rate-limiting on login/signup, and hardened session cookies — all verified with a 25-assertion automated test suite.
- Instrumented complete observability with LangSmith tracing across every agent node, giving a full node-by-node trace of each recommendation run.

End-to-End MLOps Pipeline — Phishing Website Detection

Python · Scikit-learn · Pandas · FastAPI · MongoDB Atlas · MLflow · DagsHub · Docker · AWS (EC2, S3, ECR) · GitHub Actions
[View Repository](#)

- **Designed and built the complete MLOps pipeline end to end** for a phishing-website detection system — every stage, from raw data to a live, auto-deployed API, was built and owned personally.
- Built the data ingestion layer to pull network-security data from MongoDB Atlas, followed by dedicated validation and transformation stages to clean, structure, and prepare the data before training.
- Implemented the model training and evaluation pipeline using Scikit-learn, with all experiments, parameters, and metrics logged and tracked through MLflow, integrated with DagsHub for centralized, shareable experiment history and model registry.
- Exposed the trained model as a REST API using FastAPI, and containerized the full application with Docker to make it fully portable and environment-independent.
- Automated the entire build-test-deploy cycle with GitHub Actions CI/CD: on every push, the pipeline builds the Docker image, pushes it to AWS ECR, and deploys it to an AWS EC2 instance, with AWS S3 used for artifact and model storage.
- Focused on engineering a reusable, production-style pipeline (ingestion → transformation → training → deployment → monitoring) rather than optimizing for a single accuracy number, since model accuracy naturally varies across different datasets — the value delivered here is the pipeline itself, not one fixed metric.

ACHIEVEMENTS

- ★ **8th Place, Worldwide** — **SmartReco Hackathon Build Challenge**, organized by Krish Naik Academy. Competed against participants selected from around the world and ranked 8th globally for designing, building, and deploying a fully functional agentic AI recommendation platform (SmartReco) under competition time constraints — including live LLM integration, RAG retrieval, and automated CI grading.

EDUCATION

Bachelor of Science in Data Science

Air University — Islamabad, Pakistan

CERTIFICATIONS & COURSES

- **Complete Python with DSA Bootcamp + LeetCode Exercises** — Python, Data Structures & Algorithms, Competitive Problem-Solving
- **Complete Data Science, ML, Deep Learning & NLP Bootcamp** — Machine Learning Algorithms, Deep Learning Architectures, Natural Language Processing

LANGUAGES

Urdu — Native **English** — Intermediate Professional Proficiency